

JOHANNES HORN

CURRICULUM VITAE



CONTACTS

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ACADEMIC CAREER

June 2021 - August 2026	Postdoc, University of Frankfurt, DFG-Collaborative Research Centre TRR 326 <i>Geometry and Arithmetic of Uniformized Structures</i> , project number 444845124.
Aug. 2021 and July 2022	Parental Leave.
Oct. 2020 - May 2021	Postdoc, University of Frankfurt, DFG-grant MO 1884/2-1 <i>Klassifikation von Teichmüllerkurven</i> .
Oct. 2020	Completion of Doctorate, Thesis <i>Singular fibers of Hitchin systems</i> , Karl-Ruprechts Universität Heidelberg, magna cum laude. Supervision: Prof. Dr. Anna Wienhard, Dr. Daniele Alessandrini.
Oct. 2019 - Oct. 2020	Graduate Student, Heidelberg Institute for Theoretical Studies, Research Group <i>Groups and Geometry</i> .
Aug. 2019 - Oct. 2019	Graduate Internship in Research Program <i>Holomorphic Differentials in Mathematics and Physics</i> , MSRI, Berkeley.
Feb. 2019 - Mar. 2019	Graduate Internship with Dr. Brian Collier, University of Maryland, College Park.
Oct. 2016 - Oct. 2019	Graduate Student, RTG 2229 <i>Asymptotic Invariants and Limits of Groups and Spaces</i> , University of Heidelberg.
Apr. 2015 - Oct. 2016	Master of Science Mathematics, University of Stuttgart. Thesis <i>Limiting Configurations from a Hermitian Point of View</i> , grade 1,3.
Oct. 2012 - May 2015	Bachelor of Science Mathematics, University of Stuttgart. Thesis <i>The Orthogonal Group in Characteristic 2</i> , grade 1,0.
Oct. 2011- Sept. 2012	Studies for Secondary School Teaching in mathematics, politics and economics, University of Stuttgart.
June 2009	Abitur, Ellental-Gymnasium Bietigheim-Bissingen.

THIRD PARTY FUNDS

DFG, Sachmittel Eigene Stelle, research theme “Spectral networks in Donaldson-Thomas theory”	230.336 Euro.
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PUBLICATIONS AND PREPRINTS

7. | (with Martin Möller) Wall-crossing formulas via spectral networks, 2025, [arxiv:2508.07727](#).
6. | (with Siqi He and Nianzi Li) The asymptotics of the $SL(2, \mathbb{C})$ -Hitchin metric on the singular locus: subintegrable systems, 2025, [arxiv:2506.04957](#).
5. | (with Emilio Franco, Robert Hanson and André Oliveira) Fourier-Mukai transforms and normalization of nodal curves, 2024, [arxiv:2405.11860](#).
4. | (with Johannes Schwab) Visible Lagrangians for Hitchin systems and pillowcase covers, SIGMA 22 (2026), 022, [eprint](#).
3. | (with Martin Möller) Compactifying the rank two Hitchin system via spectral data on semistable curves, Algebr. Geom. 11, No. 6, 849-894 (2024), [eprint](#).
2. | $\mathfrak{sl}(2)$ -type singular fibres of the symplectic and odd orthogonal Hitchin system, Journal of Topology, Vol. 15, Iss. 1 2022, [eprint](#).
1. | Semi-abelian spectral data for singular fibers of the $SL(2, \mathbb{C})$ -Hitchin system, Int. Math. Res. Not. (10) 2020, ISSN 1073-7928, [eprint](#).

INVITED TALKS

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| Mar. 2026 | “Wall-crossing via spectral networks”, Morningside Center of Mathematics, Chinese Academy of Science, Beijing. |
| Feb. 2026 | “From spectral networks to wall-crossing”, Winterseminar Algebra Darmstadt, Manigod. |
| June 2025 | “Approximation by semi-flat metrics for singular Hitchin fibers”, Conference Higgs bundles and harmonic maps, Christian-Albrechts-Universität zu Kiel. |
| May 2025 | “Asymptotic on the Hitchin metric on subintegrable systems”, Conference Stratifications of Higgs bundle moduli spaces and related topics, Universidade de Santiago de Compostela. |
| Apr. 2024 | “Visible Lagrangians in Hitchin systems”, Workshop: Advances in Higgs bundles, BRIN center, University of Maryland at College Park. |
| Nov. 2023 | “From Lagrangians in Hitchin systems to pillowcase covers”, Geometrieseminar, Universität Heidelberg. |
| Aug. 2023 | “Visible Lagrangians and square-tiled surfaces”, VBAC - Recent applications to the geometry of moduli spaces (contributed talk), Universität Duisburg-Essen. |

June 2022	“Singular Hitchin fibers via Hecke modifications and abelianisation”, Geometric structures (re)united (contributed talk), University of Illinois at Chicago.
May 2022	“Resolving rank 2 spectral covers by semi-stable curves”, Essener Seminar für algebraische Geometrie und Arithmetik, Universität Duisburg-Essen.
Feb. 2022	“Resolving the rank 2 Hitchin system by compactified Jacobians of semi-stable curves”, Geometria em Lisboa, Lisbon.
Dec. 2021	“Compactifying the $GL(2, \mathbb{C})$ -Hitchin system on the space of quadratic multi-scale differentials”, Workshop: Higgs bundles at infinity, Karl-Albrechts-Universität zu Kiel.
Nov. 2021	“A modified Hitchin base using quadratic multi-scale differentials”, Geometry and Topology Seminar, Universidade do Porto.
Feb. 2021	“Spectral data for singular Hitchin fibers and asymptotic solutions to the Hitchin equation”, Complex Analysis Seminar, Freie Universität Berlin.
Jan. 2021	“Singular fibers of the symplectic and odd orthogonal Hitchin system”, Geometry and Topology seminar, Universidade do Porto.
Mar. 2020	“Singular fibers of Hitchin integrable systems”, Geometry-Topology Seminar, University of Maryland, College Park.
Feb. 2020	“Singular fibers of Hitchin integrable systems”, Seminar Symplectic Geometry, Gauge Theory and Categorification, Columbia University, New York.
Nov. 2018	“Limiting configurations on the singular locus of the Hitchin map”, Oberseminar Geometrie, Karl-Albrechts-Universität zu Kiel.

SERVICE AND TEACHING

Sept 24	Coorganization of workshop <i>Ramification in geometric Langlands and non-abelian Hodge theory</i> (joint with Andreas Hohl, Judith Ludwig, Konstantin Jakob, Timo Richarz), University of Heidelberg, webpage .
ST 23 - WT 24	Representative of scientific staff (non-faculty) in faculty council, University of Frankfurt.
ST 22 - ST 23	Coorganization of Workshop series <i>Geometry of coherent sheaves: From derived categories to Higgs bundles</i> (joint with Barbara Bolognese and Inder Kaur), University of Frankfurt, webpage .
WT 24	Coorganization of GAUS-AG <i>Moduli of Quiver Representations and GIT Quotients</i> (joint with Andrei Bud and Karin Schaller), University of Frankfurt.
ST 21	Organisation of Darmstadt-Frankfurt seminar on <i>The $P=W$ conjecture</i> , webpage .
WT 22 - WT 25	Lecturing and organization of precourse for first year students of Bachelor mathematics and mathematics teacher students, University of Frankfurt, webpage .
WT 23	Colecturing of master course <i>Liegruppen und algebraische Gruppen</i> (joint with Martin Möller), University of Frankfurt, webpage .
ST 17 - WT 25	Teaching Assistance <i>Geometric Group Theory, Differential Geometry 2: Symmetric Spaces, Thermodynamic formalism, Seminar Tapetenmustergruppen</i> .

(ST: Summer Term, WT: Winter Term)

March 12, 2026